FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – I
PAPER –1
TEST DATE: 28-12-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FOUR/FIVE** statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A large insulating thick sheet of thickness 2d is charged with a uniform volume charge density ρ. A particle of mass m, carrying a charge q having a sign opposite to that of the sheet, is released from the surface of the sheet. The sheet does not offer any mechanical resistance to the motion of the particle. Find the oscillation frequency v of the particle inside the sheet:

(A) $v=\frac{1}{2\pi}\sqrt{\frac{q\rho}{m\varepsilon_0}}$ (B) $v=\frac{1}{2\pi}\sqrt{\frac{2q\rho}{m\varepsilon_0}}$

(C) $v=\frac{1}{4\pi}\sqrt{\frac{q\rho}{m\varepsilon_0}}$ (D) $\omega=\frac{1}{2\pi}\sqrt{\frac{q\rho}{m\varepsilon_0}}$

2. A charged particles with velocity $\vec{v}=x\hat{i}+y\hat{j}$ moves in a magnetic field $\vec{B}=y\hat{i}+x\hat{j}$. The magnitude of magnetic force acting on the particle is F. Which one of the following statement(s) is/are correct?

(A) No force will act on particle if $x=y$ (B) $F\propto\left(x^2-y^2\right) if\ \ x>y$

(C) The force will act along z-axis if $x>y$ (D) The force will act along y-axis if $x>y$

3. In a photoelectric effect experiment. If f is the frequency of radiations incident on the metal surface and I is the intensity of incident radiations, then which of the following is/are correct.

(A) If 'f' is increased keeping 'I' and work function constant, stopping potential and maximum kinetic energy of photoelectron increases.
(B) If distance between cathode and anode is increased, stopping potential remains same.
(C) If 'I' is increased keeping 'f' and work function constant, saturation current increases and stopping potential remains same.
(D) If the work function is decreased keeping 'f' and 'I' constant then stopping potential will increase and maximum kinetic energy of photoelectrons also increases.

4. $y\left(x,t\right)=\frac{0.6}{\left(2x-5t\right)^2+5}$ is equation of moving pulse, where x and y are in meter t is in second, then:

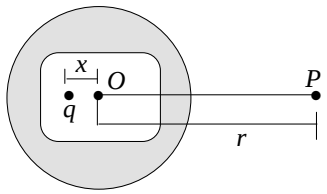
(A) Pulse is moving in +x direction (B) It is a symmetric pulse

(C) Its maximum displacement is 0.8m (D) In 1 second it will travel a distance of 2.5 m

5. The speed $\left(\text{v}\right)$ of a particle moving along a straight line, when it is at a distance $\left(x\right)$ from a fixed point on the line, is given by $v^2=144-9x^2$. Select the correct option:

(A) Displacement of the particle $\leq$ distance moved by it
(B) The magnitude of acceleration at a distance 3 units from the fixed point is 27 units
(C) The motion is simple harmonic with $\text{T}=\frac{\pi}{3}$ units
(D) The maximum displacement from the fixed point is 4 units

6. A point charge q is placed within the cavity of an electrically neutral conducting shell whose outer surface has spherical shape figure. Then:

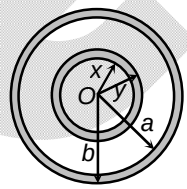


(A) The potential V at point P lying outside the shell at a distance r from the centre O of the outer surface depends upon the value of x.
(B) Potential at P does not depend upon the value of x.
(C) A total charge q will be induced on the outer surface of the shell which will be distributed uniformly on the outer surface.
(D) A total charge -q will be induced on the inner surface of the shell, which will be distributed non-uniformly on the inner surface.

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FOUR/FIVE statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. In the figure shown a conducting spherical shell of inner radius *x* and outer radius *y* is concentric with a larger conducting spherical shell of inner radius *a* and outer radius *b*. The inner shell has a total charge +3*Q* and the outer shell has a total charge +5*Q*. Let *r* be the distance of any point from the common centre *O*. Match List –I and List–II.



List-I		List-II	
(I)	Electric field strength is zero	(P)	On outer surface of the larger spherical shell
(II)	Electric field strength is non-zero	(Q)	On inner surface of the larger spherical shell
(III)	Magnitude of charge is 3*Q*	(R)	On outer surface of the smaller spherical shell
(IV)	Charge is + 8*Q*	(S)	For $a < r < b$

Which one of the following options is correct?

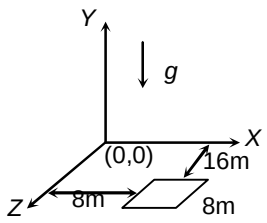
(A) $\mathrm{I} \to \mathrm{P},\ \mathrm{II} \to \mathrm{Q,R},\ \mathrm{III} \to \mathrm{Q,R},\ \mathrm{IV} \to \mathrm{R,S}$

(B) $\mathrm{I} \to \mathrm{S},\ \mathrm{II} \to \mathrm{P,Q},\ \mathrm{III} \to \mathrm{P,S},\ \mathrm{IV} \to \mathrm{R,Q}$

(C) $\mathrm{I} \to \mathrm{S},\ \mathrm{II} \to \mathrm{P,Q,R},\ \mathrm{III} \to \mathrm{Q,R},\ \mathrm{IV} \to \mathrm{P}$

(D) $\mathrm{I} \to \mathrm{Q},\ \mathrm{II} \to \mathrm{P},\ \mathrm{III} \to \mathrm{Q,S},\ \mathrm{IV} \to \mathrm{P,Q,R}$

8. A square platform of side length 8 m is situated in *x*-*z* plane such that it is at 16 m from the *x*-axis and 8 m from the *z*-axis as shown in figure. A particle is projected with velocity $\vec{v} = (v_2\hat{i} + 25\hat{j})$m/s relative to wind from origin and at the same instant the platform starts with acceleration $\vec{a} = (2\hat{i} + 2.5\hat{j})$m/s^2. Wind is blowing with velocity $v_1\hat{k}$. (g = 10 m/s^2)



List-I		List-II	
(I)	Possible values of v_2 (in m/s) so that particle hits the platform is/are	(P)	4
(II)	Possible value of v_1 (in m/s) so that particle hits the platform is/are	(Q)	6
(III)	Possible time of flight (in second) of the particle is/are	(R)	8
(IV)	Possible value of displacement/s (in m) of the particle in *y*-direction when v_2 has its minimum possible value is/are (till particle hit the platform)	(S)	20

Which one of the following options is correct?

(A) $\text{I} \to \text{Q,R, II} \to \text{P,Q, III} \to \text{P, IV} \to \text{S}$

(B) $\text{I} \to \text{Q, II} \to \text{Q, III} \to \text{P, IV} \to \text{S}$

(C) $\text{I} \to \text{S, II} \to \text{P,Q, III} \to \text{Q,R, IV} \to \text{P,S}$

(D) $\text{I} \to \text{Q,R, II} \to \text{P, III} \to \text{Q,S, IV} \to \text{P,Q,R}$

9. Match the statements in List–I with the results in List–II.

List-I		List-II	
(I)	A thin uniform spherical shell of surface area *S* has an initial temperature more than its surrounding atmosphere. Then magnitude of rate of change of its temperature with time	(P)	is independent of *S*
(II)	A soap bubble initially in equilibrium is given a charge *Q*, which distributes uniformly over its surface. The centre of the bubble is always fixed. For the duration the bubble having surface area *S* expands, the magnitude of electric potential at a fixed point always lying outside the bubble.	(Q)	depends on *S*
(III)	A container with open top and filled with ideal liquid is placed at rest on a smooth horizontal table. A small hole of area *S* is drilled at the bottom of a side wall of container. The magnitude of force exerted by escaping liquid on the container.	(R)	remains constant
(IV)	An infinitely long straight current carrying wire lies along the axis of a closed cylindrical surface of total surface area *S* in space. As the magnitude of current in the wire is continuously increased, the magnitude of the magnetic flux through the surface of this cylinder.	(S)	decreases with time

Which one of the following options is correct?

(A) $\text{I} \to \text{S, II} \to \text{P,Q, III} \to \text{R,S, IV} \to \text{P,R}$

(B) $\text{I} \to \text{P,S, II} \to \text{P,R, III} \to \text{Q,S, IV} \to \text{P,R}$

(C) $\text{I} \to \text{S,R, II} \to \text{R, III} \to \text{Q,S, IV} \to \text{R,S}$

(D) $\text{I} \to \text{S, II} \to \text{P,R, III} \to \text{P,Q, IV} \to \text{Q,R}$

10. Related to photoelectric effect, in List-I, some physical quantities change while in List-II effects of these changes are given. Match the entries of List-I with the entries of List-II.

List-I		List-II	
(I)	Intensity of incident light changes	(P)	K_{max} of emitted photoelectrons changes
(II)	Frequency of incident light changes	(Q)	Stopping potential changes
(III)	Target material changes	(R)	Saturation current changes
(IV)	Potential difference between the emitter and collector changes	(S)	Time delay in emission of photoelectrons changes

Which one of the following options is correct?

(A) $I \rightarrow R,\ II \rightarrow P,\ III \rightarrow Q,S,\ IV \rightarrow S$

(B) $I \rightarrow S,\ II \rightarrow Q,S,\ III \rightarrow P,Q,\ IV \rightarrow S$

(C) $I \rightarrow R,\ II \rightarrow P,Q,\ III \rightarrow P,Q,\ IV \rightarrow S$

(D) $I \rightarrow S,\ II \rightarrow P,Q,\ III \rightarrow Q,S,\ IV \rightarrow R$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

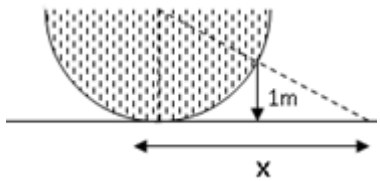
11. If resistance R_1 in resistance box is 300Ω, then the balanced length is found to be $75.0\,cm$ from end A. The diameter of unknown wire is 1 mm and length of the unknown wire is $31.4\,cm$. Find the specific resistance of the unknown wire $(\mu\Omega\text{-m})$

12. A slab of glass of thickness 6cm and index 1.5 is placed somewhere in between a concave mirror and a point object, perpendicular to the mirror's optical axis. The radius of curvature of the mirror is 40 cm. If the reflected final image coincides with the object, then find the distance of the object from the mirror (in cm).

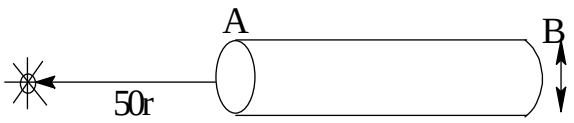
13. A terrorist places a bomb at a horizontal distance of 6m from the foot of building of height 8m. When the bomb explodes its fragments fly in all directions with a velocity upto 30 m/s. Find how long (in seconds) a man on the top of the building will be in danger. ($g=10\ ms^{-2}$)

14. A string is attached 10a horizontal cylinder of radius r = 4cm around which it passes several times. The free end hangs vertically and supports a particle. The vertical portion is equal to $l_0 = 28cm$. The particle is given' a horizontal velocity u perpendicular to the axis of the cylinder and in such a sense as to wind up the string around the cylinder. What is the value of u (in m/sec) so that the string does not slacken in the subsequent motion?

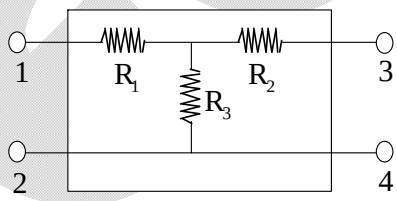
15. A hemispherical tank of diameter 4m is filled with water. A very 'small hole is punched at 1m above the bottom as shown in figure. Find x (in m) (distance at which water strikes the surface).



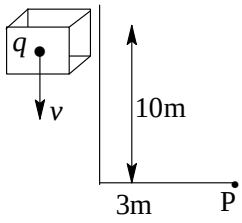
16. A cylindrical rod of length 64 cm and cross-section radius $r=\left(2/\sqrt{\pi}\right)\text{cm}$ is placed at a distance $50\,r$ from a infrared point source S of power 1.25 kW as shown in the figure. The lateral surface of the rod is perfectly insulated from the surroundings. The cross-section A absorbs 80% of the incident energy, has temperature T_A in steady state. The surface B is radiating energy into space and the wavelength emitted by it with maximum energy density is $100,000\ \text{Å}$. Find the value of T_A (temperature of end A in Kelvin) if conductivity varies with temp as $\text{K}=\frac{\text{T}}{\text{T}_\text{A}}$. Assume that the rate of flow of heat through the rod is steady.
(Wein's constant $=0.003\ \text{K}$)



17. A concealed circuit (a black box) consisting of resistors has four terminals. If a voltage is applied between clamps 1 and 2 when clamps 3 and 4 are open, the power liberated is $P_1=40\ \text{W}$ and when clamps 3 and 4 are closed, the power liberated is $P_2=80\ \text{W}$. If the same source is connected to the clamps 3 and 4, the power liberated in the circuit when clamps 1 and 2 are open is $P_3=20\ \text{W}$. Determine the power P_4 (in Watt) consumed in the circuit when the clamps 1 and 2 are connected and the same voltage is applied between the clamps 3 and 4.



18. An elevator carrying a charge of 0.2 C, is moving down with a velocity of $4\times10^3\ \text{m/s}$. The elevator is 10 m from the bottom and 3 m horizontally from P as shown. What magnetic field (in μT) does it produces at point P



Chemistry

PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

19. Which of the following statement (s) is/are correct?
(A) Fe^{3+} and Mn^{2+} have equal paramagnetic character
(B) Cu_2Cl_2 and $CuCl_2$ are coloured.
(C) MnO_4^- is purple in colour because of unpair d electrons
(D) The magnetic moment of Fe^{2+} and Co^{3+} both are equal to $2\sqrt{6}$ B. M.

20. Which of the following statements are correct?
(A) The total spectral lines obtained from a single line during Zeeman effect is $(2l + 1)$.
(B) In the Lyman series as the energy liberated during transition increases then the distance between the spectral lines goes on decreasing.
(C) The highest probability of finding an electron in 1s orbital is in the vicinity of the circumference.
(D) The highest probability of finding an electron in 1s orbital is exactly at the middle between nucleus and circumference.

21. Which of the following is/are correct?
(A) Ammonium salts are more soluble than sodium salts
(B) $SnCl_2$ is more ionic than $SnCl_4$
(C) Calcium fluoride is more ionic than calcium chloride
(D) The formal charge on S atom in SO_2 is four

22. Which of the following statements are true for a gas -
(A) The mean free path is inversely proportional to the pressure of the gas at constant temperature
(B) The mean free path is proportional to temperature at constant pressure
(C) The collision frequency among the molecules of gas is proportional to temperature
(D) The velocity possessed by the largest fraction of molecules at a given temperature is known as root mean square velocity

23. Which of the following statements is/are false -
(A) Endothermic reactions are never spontaneous, at any temperature
(B) Exothermic reactions are always spontaneous at any temperature
(C) Exothermic reactions in which the entropy change for the system is negative are spontaneous at any temperature
(D) Reactions in which the entropy change for the system is negative are never spontaneous at any temperature

24. An aqueous solution contains either Hg_2^{2+} or Hg^{2+} the given solution gives green ppt with KI solution but no precipitate with H_2S. About the given aqueous solution which of following is correct -
(A) It contains Hg_2^{2+}
(B) It contains Hg^{2+}
(C) With NH_3 solution it gives black precipitate
(D) None of these

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FOUR/FIVE statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the following:

List-I		List-II	
(I)	O < Te < Se < S	(P)	Acidic character of oxides
(II)	$Be^+ < C^+ < B^+ < N^+ < F^+ < O^+ < Li^+$	(Q)	Electron Affinity
(III)	$Fe^{2+} < Fe^{3+} < Fe^{6+}$	(R)	Atomic radius
(IV)	N < P < As < Sb	(S)	Ionisation energy

Which one of the following options is correct?

(A) $I \rightarrow P,\ II \rightarrow R,\ III \rightarrow S,\ IV \rightarrow R$

(B) $I \rightarrow Q,\ II \rightarrow S,\ III \rightarrow P,Q,S,\ IV \rightarrow R,Q$

(C) $I \rightarrow S,\ II \rightarrow P,\ III \rightarrow Q,R,\ IV \rightarrow P,S$

(D) $I \rightarrow Q,\ II \rightarrow P,\ III \rightarrow Q,S,\ IV \rightarrow P,Q,R$

26. Match the following:

List-I		List-II	
(I)	$XeOF_4$	(P)	no lone pairs of electron
(II)	XeO_4	(Q)	one lone pair of electrons.
(III)	XeF_4	(R)	more than one lone pairs of electrons
(IV)	IF_5	(S)	non planar

(A) $I \rightarrow Q,S,\ II \rightarrow P,\ S,\ III \rightarrow R,\ IV \rightarrow Q,S$

(B) $I \rightarrow Q,\ II \rightarrow S,\ III \rightarrow P,Q,S,\ IV \rightarrow R,Q$

(C) $I \rightarrow S,\ II \rightarrow P,\ III \rightarrow Q,R,\ IV \rightarrow P,S$

(D) $I \rightarrow Q,\ II \rightarrow P,\ III \rightarrow Q,S,\ IV \rightarrow P,Q,R$

27. Match the following:

List-I		List-II	
(I)	$\left[CoCl_3\left(NH_3\right)_3\right]$	(P)	show facial form
(II)	$\left[Cr\left(Ox\right)_3\right]^{3-}$	(Q)	cis form is optically active
(III)	$\left[CrCl_2\left(Ox\right)_2\right]$	(R)	Transform is optically inactive
(IV)	$\left[RhCl_3\left(Py\right)_3\right]$	(S)	show meridian form

(A) $I \rightarrow Q,S,\ II \rightarrow P,\ S,\ III \rightarrow R,\ IV \rightarrow Q,S$

(B) $I \rightarrow Q,\ II \rightarrow S,\ III \rightarrow P,Q,S,\ IV \rightarrow R,Q$

(C) $I \rightarrow S,\ II \rightarrow P,\ III \rightarrow Q,R,S,\ IV \rightarrow P,S$

(D) $I \rightarrow P,S,\ II \rightarrow Q,\ III \rightarrow Q,R,\ IV \rightarrow P,S$

28. Match the following:

List-I		List-II	
(I)	Paramagnetic	(P)	N_2O_3
(II)	Dark blue liquid in liquid or solid state	(Q)	NO_2
(III)	Unstable gas due to auto decomposition.	(R)	NO
(IV)	Brown gas	(S)	N_2O_5

(A) $I \rightarrow Q,S,\ II \rightarrow P,S,\ III \rightarrow R,\ IV \rightarrow Q,S$

(B) $I \rightarrow Q,\ II \rightarrow S,\ III \rightarrow P,Q,S,\ IV \rightarrow P,S$

(C) $I \rightarrow Q,R,\ II \rightarrow P,\ III \rightarrow P,S,\ IV \rightarrow Q$

(D) $I \rightarrow P,S,\ II \rightarrow Q,\ III \rightarrow Q,R,\ IV \rightarrow P,S$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. A mixture of CH_4 and C_2H_2 occupied a certain volume at a total pressure equal to 63 torr. The same gas mixture was burnt to CO_2 and H_2O (ℓ). The CO_2 (g) alone was collected in the same volume and at the same temperature, the pressure was found to be 69 torr. What was the mole fraction of CH_4 in the original gas mixture?

30. N_2 gas is assumed to behave ideally. A given volume of N_2 originally at 373 K and 0.1013 MPa pressure is adiabatically compressed due to which its temperature rises to 673 K. What is final pressure (in MPa) of gas?

31. A saturated solution of iodine in water is 1.25×10^{-3} (M). In any saturated solution of I_2 concentration of I_2 is 1.25×10^{-3} (M). In 1 L of 0.1 (M) solution I^-, it is seen 51.25×10^{-3} mole of I_2 can be maximum dissolved. In the aqueous solution of I^-(aq), I_2 (ag) undergoes complex formation, I_2 (aq) + $I^- \underset{}{\overset{K_C}{\rightleftharpoons}} I_3^-$ (aq).
What is the value of K_C?

32. The reaction, N_2O_5 (g) $\rightleftharpoons$ 2 NO_2 (g) + $\frac{1}{2}$ O_2 (g), is started with initial pressure of N_2O_5 (g) equal to 600 torr. What fraction of N_2O_5 (g) decomposed when total pressure of the system is 960 torr ?

33. At 298 K, the conductivity of a saturated solution of AgCl in water is 2.6×10^{-6} ohm^{-1} cm^{-1}. Given, λ_m^∞ (Ag^+) = 63 ohm^{-1} cm^2 mol^{-1} & λ_m^∞ (Cl^-) = 67 ohm^{-1} cm^2 mol^{-1}
The solubility product of AgCl is $x \times 10^{-10}$, then value of 'x' is ____.

34. For a certain reaction of order n, the time for half change, $t_{1/2}$, is given by $t_{1/2} = \frac{[2-\sqrt{2}]}{k} \times C_0^{1/2}$ where k is the rate constant and C_0 is the initial concentration what is n ____.

35. The density of solid argon (Ar = 40 g/mol) is 1.68 g/mL at 40 K. If the argon atom is assumed to be a sphere of radius = 1.50×10^{-8} cm, what % of solid Ar is apparently empty space?

36. Two liquids X and Y form are ideal solution. At 300 K, vapour pressure of the solution containing 1 mole of X and 3 moles of Y is 550 mm Hg. At the same temperature, if 1 mole of Y is further added to this solution, vapour pressure of the solution increases by 10 mm Hg. What is the sum of vapour pressure (in mm Hg) of X and Y in their pure states?

Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. Let A and B are commutative square matrices of order three such that A is symmetric and B is skew symmetric, if $C=(A+B)^T(A+B)(A-B)^{-1}$ (where X^T and $|x|$ denotes transpose and determinant value of x respectively), then

(A) $A+B-C=0$ (B) $A-B-C=0$

(C) $|B|=0$ (D) $2A-C=C^T$

38. If $\lim\limits_{n\to\infty}\left(\frac{1}{1^2}+\frac{1}{2^2}+.....+\frac{1}{n^2}\right)=\frac{\pi^2}{6}$ and $S=\lim\limits_{n\to\infty}\left(\frac{1}{1^3\cdot 2^3}+\frac{1}{2^3\cdot 3^3}+\frac{1}{3^3\cdot 4^3}+....\frac{1}{n^3\cdot(n+1)^3}\right)$, then $S+\pi^2$ is equal to

(A) A non-integral number (B) An rational number

(C) A transcendental number (D) An integer

39. $f(x)$ is a differentiable function such that $f(x)=x^2+\int_0^x e^{-t}f(x-t)dt$, then

(A) $f(x)=\frac{x^3}{3}+x$ (B) $f(x)=\frac{x^3}{3}+x^2$

(C) $f'(x)=x^2+2x$ (D) $\int_{-2}^{2}\left(f(x)-x^2\right)dx=0$

40. Mr. A randomly picks 3 distinct numbers from the set $\{1,2,3,.....,9\}$ and arranges them in descending order to form a three digit number. Mr. B randomly picks 3 distinct numbers from the set $\{1,2,3,....,8\}$ and also arranges them in descending order to form 3 digit numbers.

(A) Probability that Mr. A's 3 digit number is always greater then Mr. B's 3 digit number is $\frac{1}{3}$

(B) Probability that Mr. A and Mr. B has the same 3 digit numbers is $\frac{1}{84}$

(C) Probability that Mr. A's 3 digit number is greater then Mr.B's 3 digit number is $\frac{37}{56}$

(D) Probability that Mr. A's 3 digit number is greater then Mr.B's 3 digit number is $\frac{1}{3}$

41. Let $H:\frac{(x-1)^2}{9}-\frac{(y+2)^2}{4}=1$ be the hyperbola with centre at C. Now if from any point 'Q' on the asymptotes a straight line be drawn perpendicular to the transverse axis to meet the hyperbola at P and P'. Again a tangent is drawn to the hyperbola at P which meet the asymptotes at L and G, then

(A) $PQ.QP'=4$ (B) $PQ.QP'=9$

(C) area of $\Delta CLG=6$ (D) area of $\Delta CLG=8$

42. If $I_1=\int_0^{\frac{\pi}{2}}\frac{\sin(\sin x)}{\sin x}dx; I_2=\int_0^{\frac{\pi}{2}}\frac{\sin x}{x}dx; I_3=\int_0^{\frac{\pi}{2}}\frac{\sin(\tan x)}{\tan x}dx$, then which of the following is true

(A) $I_1>I_3$ (B) $I_2>I_3$

(C) $I_1>I_2$ (D) $I_1<I_2$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FOUR/FIVE statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Let $f(x)=\begin{cases}(15-3b)\{x\}-(b^2-4b-5)\operatorname{sgn}(x+1), & \frac{-\pi}{2}<x<0\\ k([x]+[-x]), & 0\le x\le\pi\\ \frac{(a+2\cos x)(1+\tan x)}{\ln(1+\pi^2-2\pi x+x^2)}, & \pi<x<\frac{3\pi}{2}\end{cases}$

Where $[y]$, $\{y\}$ and $\operatorname{sgn}(y)$ denote greatest integer function, fractional part function and signum function of y respectively.

List-I		List-II	
(I)	If f is continuous in $\left(\frac{-\pi}{2},0\right)$, then the value of b is	(P)	0
(II)	If f is continuous at $x=\pi$, then value of $(a+k)$ is	(Q)	1
(III)	If f is continuous in $\left(\frac{-\pi}{2},\pi\right)$, then value of $(b+k)$ is	(R)	5
(IV)	If f has exactly four points of discontinuity in $\left(\frac{-\pi}{2},\frac{3\pi}{2}\right)$, then $(a+b+k)$ is equal to	(S)	6

Which one of the following options is correct?

(A) $I\to R,\ II\to Q,\ III\to R,\ IV\to S$

(B) $I\to P,\ II\to Q,\ III\to R,\ IV\to S$

(C) $I\to S,\ II\to P,\ III\to Q,R,\ IV\to P$

(D) $I\to Q,\ II\to P,\ III\to Q,S,\ IV\to P$

44. Match the following:

List-I		List-II	
(I)	The number of solution(s) of the equation $\sin^{-1}\left(\frac{1}{1+x^2}\right)=\operatorname{sgn}\left(x^2+1\right)$, is (are)	(P)	4
(II)	If the range of the function $f(x)=\cos^{-1}\left(\frac{x^2}{1+x^2}\right)$ is $(a,b]$, then $[a+b]$ is equal to	(Q)	1
(III)	If α and β are the roots of the $2x^2-3x-2=0$, then $\frac{12}{17}\left(\tan^2\left(\sin^{-1}\frac{\alpha}{\sqrt{1+\alpha^2}}\right)+\tan^2\left(\cos^{-1}\frac{1}{\sqrt{1+\beta^2}}\right)\right)$ is	(R)	2
(IV)	If $f(x)=\sin\left(\cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right)+\sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)$, then the value of $\sum_{x=1}^{4} f\left(\frac{3x}{16}\right)$ is equal to	(S)	3

Which one of the following options is correct?

(A) $\mathrm{I}\to\mathrm{P},\ \mathrm{II}\to\mathrm{R},\ \mathrm{III}\to\mathrm{S},\ \mathrm{IV}\to\mathrm{R}$

(B) $\mathrm{I}\to\mathrm{R},\ \mathrm{II}\to\mathrm{Q},\ \mathrm{III}\to\mathrm{S},\ \mathrm{IV}\to\mathrm{P}$

(C) $\mathrm{I}\to\mathrm{S},\ \mathrm{II}\to\mathrm{P},\ \mathrm{III}\to\mathrm{R},\ \mathrm{IV}\to\mathrm{P}$

(D) $\mathrm{I}\to\mathrm{Q},\ \mathrm{II}\to\mathrm{P},\ \mathrm{III}\to\mathrm{Q},\ \mathrm{IV}\to\mathrm{R}$

45. Consider a system of linear equation $3x+y-z=0,\ x-\frac{py}{4}+z=2$ and $2x-y+2z=q$ where $p,q\in I$ and $p,\ q\in[1,10]$, then identify the correct statement(s).

List-I		List-II	
(I)	Number of ordered pairs (p,q) for which system of equation has unique solution is	(P)	1
(II)	Number of ordered pairs (p,q) for which system of equation has no solution is	(Q)	9
(III)	Number of ordered pairs (p,q) for which system of equation has infinite solution is	(R)	91
(IV)	Number of ordered pairs (p,q) for which system of equation has atleast one solution is	(S)	90

Which one of the following options is correct?

(A) $\mathrm{I}\to\mathrm{P},\ \mathrm{II}\to\mathrm{R},\ \mathrm{III}\to\mathrm{S},\ \mathrm{IV}\to\mathrm{R}$

(B) $\mathrm{I}\to\mathrm{Q},\ \mathrm{II}\to\mathrm{S},\ \mathrm{III}\to\mathrm{P},\ \mathrm{IV}\to\mathrm{R}$

(C) $\mathrm{I}\to\mathrm{S},\ \mathrm{II}\to\mathrm{Q},\ \mathrm{III}\to\mathrm{P},\ \mathrm{IV}\to\mathrm{R}$

(D) $\mathrm{I}\to\mathrm{Q},\ \mathrm{II}\to\mathrm{P},\ \mathrm{III}\to\mathrm{S},\ \mathrm{IV}\to\mathrm{P}$

46. March the following:

List-I		List-II	
(I)	If $\int \frac{\cos x-\sin x+1-x}{e^x+\sin x+x}dx=\ln\left(h\left(x\right)\right)+g\left(x\right)+c$, where c is arbitrary constant and $f\left(x\right)=h\left(x\right)+g\left(x\right)$, then	(P)	$f'(0)=3$
(II)	If $f\left(x\right)$ is a quadratic polynomial such that $f\left(0\right)=1$ and $\int \frac{f\left(x\right)}{x^2\left(x+1\right)^3}dx$ is a rational function, then	(Q)	$f'(0)=2$
(III)	If $\int \frac{\cos x\left(1+4\cos 2x\right)}{\sin x+4\sin x\cos^2 x}dx=\frac{1}{2}\ln\left\lvert h\left(x\right)\right\rvert+\frac{1}{2}\ln\left\lvert f\left(x\right)\right\rvert+c$, then	(R)	$f\left(x\right)$ is neither even nor odd function
(IV)	If $\int \frac{x^4+1}{x\left(x^2+1\right)^2}dx=A\ln\left\lvert x\right\rvert+\frac{B}{1+x^2}+c$, where c is arbitrary constant and $f\left(x\right)=Ax^2+2Bx$, then	(S)	$f\left(0\right)=0$

Which one of the following options is correct?

(A) $\mathrm{I}\rightarrow \mathrm{P,R},\ \mathrm{II}\rightarrow \mathrm{Q},\ \mathrm{III}\rightarrow \mathrm{R},\ \mathrm{IV}\rightarrow \mathrm{S}$

(B) $\mathrm{I}\rightarrow \mathrm{Q},\ \mathrm{II}\rightarrow \mathrm{S},\ \mathrm{III}\rightarrow \mathrm{P,Q},\ \mathrm{IV}\rightarrow \mathrm{R,S}$

(C) $\mathrm{I}\rightarrow \mathrm{S},\ \mathrm{II}\rightarrow \mathrm{Q,R,S}\ \mathrm{III}\rightarrow \mathrm{P},\ \mathrm{IV}\rightarrow \mathrm{P,S}$

(D) $\mathrm{I}\rightarrow \mathrm{Q,R},\ \mathrm{II}\rightarrow \mathrm{P,Q},\ \mathrm{III}\rightarrow \mathrm{S},\ \mathrm{IV}\rightarrow \mathrm{Q,R,S}$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. A triangle is so placed that the midpoints of its sides one placed on the co-ordinate axes, if a, b and c are the sides of the triangle, then the equation of the triangle is $\frac{x}{x_1}+\frac{y}{y_1}+\frac{z}{z_1}=1$ where $a^2+b^2+c^2=k\left(x_1^2+y_1^2+z_1^2\right)$, then the value of k is_____

48. Let f be real function defined on R (the set of real numbers) such that $\mathrm{f'(x)}=100\ \left(\mathrm{x}-1\right)\left(\mathrm{x}-2\right)^2(x-3)^3.....(x-100)^{100}$, for all $x\in R$. If g is a function defined on R such that $\int_a^x e^{f(t)}dt=\int_0^x g(x-t)dt+2x+3$, If some of the all the values of x for which g(x) has a local extremum be λ then find $\frac{\lambda}{3}$

49. If $\lim_{x\rightarrow 1}\frac{100}{1-x^{100}}-\frac{50}{1-x^{50}}=5\mathrm{A}$; find the value of A.

50. If S_1, S_2, S_3 denote the sums of first twenty terms of three non constant sequence in A.P., whose first terms are unity and common differences are in H.P. Then $\frac{2S_3S_1 - S_1S_2 - S_2S_3}{S_1 - 2S_2 + S_3}$ is equal to___

51. If P is a point $(2,4)$ on the parabola $y^2 = 8x$ and PQ is a focal chord, the coordinate of the mirror image of Q with respect to tangent at P are given by (α, β) then $(\beta - \alpha)$ is_____

52. If sum of the roots of the equation $\sqrt{2}\sin\left(x + \frac{\pi}{4}\right)\left(\sec^2 x - \left(1+\sqrt{3}\right)\tan x + 2 + \sqrt{3}\right)$ $= \sqrt{9 + 18\sin x.\cos x}$ in $[0, 2\pi]$ is $\frac{p}{q}\pi$ where $p, q \in N$, then find the least value of $|p - q|$

53. Let S be infinite sum of the series $2 + 3\cos x + 4\cos^2 x + 5\cos^3 x + ...\infty$, where x satisfies the equation $|5\cos x + 4| + |5\cos x - 2| = 6$. If the least value of S is equal to $\left(\frac{a}{b}\right)$ where a and b are co-prime numbers, then the value of $(a+b)$ is:

54. If the coefficient of x^8 in the expansion of $\left(2 + 9x^3 + 6x^4 + x^5\right)^{10}$ is $5.2^p.3^q$ where $p, q \in N$, then find the value of $(p+q)$